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Mail from CMU-10A rcvd at 23-JUL-76 1227-PDT
From: ALLEN NEWELL(A310AN02) at CMUA
Subject: Comments on thesis
- - - -
COMMENTS ON DOUG LENAT'S THESIS
File LENAT.THS 23 Jul 76 Newell

GENERAL COMMENTS

> It is acceptable -- hence, I accept it

> As a piece of scientific work
> Actually, it is downright significant
> But the purpose of this document is not to evaluate generally but to indicate where there are difficulties that need to be responded to
> So, little positive reinforcement herein!
> Though DL must decide for himself which of the remarks he wishes to respond to -- if any

> As a document, basically
> Again, I will make a number of comments, but I don't much care whether you take them into account or not

> What I missed most

> A better feeling for the structure of the task environment (TE)
> Some feeling comes through as a fixed structure in which AM is undergoing a fixed passage down a ravine of fixed structure
> But a better attempt to assess this structure would have been nice
> I really don't expect anything to be done on this

> Some sort of integrative display of the behavior that would have let me sense what was going on
> I am not unhappy per se with the listing of the moves in a linear string -- it just didn't do the whole job
> This is especially true given that the behavior, being multiprocessed, is discontinuous
> So that the linear stream is especially non-revealing
> This is, of course, noted
> How about a graph in concept-space which we see operations linked onto the items they depend on .
> Labeled by time, so can see how it jumps around

> An analysis of the nature of the knowledge held in the heuristics
> At least there is a summary on pages 132-133
> But understanding this heuristic knowledge rather than treating it just as if it were one huge list of specific items (each of which can of course be discussed and is) would be a real advance
> Again, I don't expect anything to be done along this line

> The analysis of the operation of the system by means of simplified performance models

> An analysis of the lack of directionality in AM under

various conditons with respect to the possibilities for goal oriented activity
- > There were some of clues lying around
  - > The alternation of wandering with spurts of directed activity
    - > As a side note, GPS behaved just this way
  - > The attempt to put in additional focussing mechanism, such as "focus of attention"
  - > The present scheme, if analysed, was seen to be a one-step set of pulses
- > One could have introduced (or analyzed the introduction of) setting some mini-goals that would make the system stay on the track of conjectured generalizations etc for several steps
  - > Instead it packages all such multi-step goals as macro-moves
- > Thus, the present system would have been seen not to be radically opposed to a goal directed system (as in all the remarks about AMs unique structure), but simply a system that only had a partial set of mechanism for complete reasoning about its domain
- > An the system would have had the neat property of conjecturing its own goals
  - > So we finally would have had an example system which started out without goals (but with goal generators)
    - > And hence would have had no top goal

> What I thought was overdrawn

- > That AM was not a theorem prover
  - > That just depends on the interpretation
    - > I mean that rather strictly
    - > Whether X is a theorem prover is a question of how one interprets its operators
    - > Structurally it is just a HS program
  - > But even more closely, consider a semantically oriented prover that established its theorems by (infinite) iteration through the instances of a model
    - > Then a program that iterates only through N instances and rejects if false, is an approximation to a theorem prover
    - > And the fact that it can make errors is pretty irrelevant to the nature of its operation
      - > Since all theorem provers have some probability of error associated with them in any event
  - > More interesting would have been the understanding of the structure of the inferences to see in what way it might really differ
    - > For instance, the operators on the surface all seem unary -- they take a single concept as input
      - > But all substantial logical systems have at least a binary inference rule (A, A implies B => B), which produces a sort of sexual evolution, rather than a merely asexual one
        - > Ie, it get combinatorial recomination
    - > I didn't see operators that ever laid their hands on two or more concepts to produce some new thing

- > The emphasis on rippling
  - > Which seems to me pretty obvious

    > Surely worth noting and surely worth writing down the
      recursive equations on the concepts which permitted the
      various search operations
    > But this sort of thing is pretty clear since SIR and GRAIS

> What stylistic features concerned me

  > The continuous attempts at being a bit cute sometimes go awry
    > I heavy (ie long and tedious) technical material a certain
      amount of enlivenment is a great thing
      > So I am not against it in principle
    > But the touch is fair ways from sure

  > The thesis appeared not to be written for other AI scientists
    > Rather, for a lay audience of some kind
    > The treatment on rippling is probably a consequence of this
    > But there is much else that reflects it
    > I prefer my science written by scientists for peer scientists
      > And let the attempt to popularize it come later
      > And I don't mean obscure or jargony, I just mean straight
        and with an attempt to be as clear about it as possible
    > I am not really upset about this; just stating preferences

> A special concern

  > AM has lots of capability -- it has it built into the use
    use of evaluation formulas etc
    > Eg, it can take square roots

  > But the use of these capabilities is carefully shielded from
    other parts of the system
    > Ie, AM is made not to know these things as far as concepts

  > Thus AM is in some sense artificially ignorant

  > Said better perhaps, AM doesn't show that a system that really
    doesn't have the concepts can get them
    > Since AM does have and use the concepts in part

  > Actually, this is a version of the Accessibility problem
    > Of packaging knowledge up in procedures in a way that the
      the program can't get at this knowledge for other purposes

> The thesis is not too good on understanding the relation of
  AM to other efforts and work in the field

  > I suppose its because DL thinks of the program as sui generis
    in many ways
    > Which it is, but not in all the ways he thinks I would guess
    > However, he does deal strongly with HS
      > But not well with TPers and Game Players

  > But there have been other programs to discover interesting
    theorems
    > Wang did some experiments with his theorem prover
      > Reported in one of the papers (in IBMJ, I think)
    > Pitrat did a whole thesis on this
      > I believe HAS loaned him a copy of this (in French)

  > What about Saul Amarel's early work on theory formation?

> Actually it would have been interesting to discuss the task that Amarel set for his system in terms of AM

> The program bears some kinship to various induction programs
> > Dendral, Sequence extrapolation, PAS-II, Analogy programs

> What about Fogel's Simulation of Intelligence through Evolution (book)

DETAILED COMMENTS

> Pretty randomly destributed, but for what they are worth
> > I will count lines strictly from the top or bottom, including all headings and equations etc
> > > Except the "Draft: Not for .." line, and footnotes

> p1 l15. Not sure you delivered on "unexpected" powers & limitations

> p4 l-11. It surely could not have been otherwise, given they little development that was possible compared to the starting concepts

> p7 l-11. What's the big deal about agendas? That is the way everyone manages a best-first search
> > How else would you do it, except to create a problem list ordered by the best-evaluation
> > > Simply processing considerations lead one to put the elements on it order, so rule is "take first"

> p7 l-3. False. Or rather, you again confuse interpretation with structure
> > Your operators are operators. Who care whether you call them "heuristic" or "legal"?

> p8 l5. The property in question is common to all mathematics
> > A theorem is proved (it is an object) and immediately comes available to be used in other proofs (as operator)
> > Thus all theorem provers have this property
> > > Thus the duality between "implies" and "deduces" which are essentially the same, one in the object (to be applied by a generalized rule of inference) and the other as a statement of what a rule of inference does
> > > Again, this is why inference systems take two inputs
> > > > R(A,A implies B) => B is just Q[A implies B](A) => B where Q[A implies B] is a constructed operator

> p8 fn17. A thought: I would have like to see some null models developed, ie, simple G&T structures that attempted to produce the sorts of conjectures by much more trivial means
> > Given that interesting concepts are simple expressions in the right language, there might be some surprises down this way

> p11 fn1. A real nit: I really prefer "Nbr"to "No", which clashes to strongly with negation for me

> p14 l2. Well, I'm not so sure. Your "symbolic reasons" are not symbolic at all, they are just a set of features which takes on numerical values
> > Looks very like an evaluation function is a game playing program
> > With lots of points for everything in sight

> I was, in fact, hoping for more
> Though I agree that the insensitivity of the behavior to details of evaluation makes it not urgent to explore this aspect

> p14 fn3. Why? Consider how many irrelevant activities a research engages in while thinking -- puff on cigarate, doodle on page, look out window, ...

> p18 "Notice the skip ..." should come after "** Task 68 **

> p25 l8. It is true of all HS programs that have a branchiness >1 that they will never explore most tasks. AM is not unique in this
> It is true that the branchiness is very large compared to a number of other HSers

> p25 l11. Good point. I was looking for what fraction was spent on decision time vs action time, and never found it -- interesting number
> See the argument Ch14 HPS

> p25 l-8. Odd statement. This says, interpreted strictly, that a factor of 10 or 100 in speed is all you need.
> After all the KI10 is a relatively slow machine

> p25 l-8 By the way, the attitude expressed here and elsewhere that it is the impression on the user interacting in real time with AM that counts strikes me as a false note
> What does that say about ELIZA?

> p25 l-5. You are setting straw men up here a bit. The "favoring" such as it is, has nothing to do with machines, only with the slow growth of our understanding how to be sophisticated
> Also, see CAPS (Berliner) and even see NSS program in 58
> Have you ever examined the evaluation functions of, say, CHESS-4.0 in detail. There are a lot of numbers around

> p27 "3.3.1 ..." Again, what is big deal about agendas? LT had one just like yours in 1956, and many after it
> Depth-first-ers can avoid it, but few else
> It does seem, however, that your set up has a strong sort of multiprocessing flavor
> I can't tell whether that is just the way we talk about it now or what
> In this respect your system has very much the flavor of HEARSAY-II
> In which huge numbers of instantiations of tasks get built up and put on the task list
> Though they have some dynamic shifting of scores which causes dynamic reordering of the task list

> p28 p10. I haven't checked, but I feel like you are getting pretty damn redundent about stating the basic cycle
> Hey, here it is again on p29 l9!

> p29 l-13 Hardly unique to Hewitt. See, eg, Normon and Bobrow's article on data limited and resource-limited processing -- or a whole book by Kahneman built on the idea of distribution of limited processing effort.

> p33 fn. Seriously, you are probably wrong. The number would still be .05, the base would be 9

> p34 l-7. Here is one of the few places where you fall into the standard trap of saying that it is trival to do something which in fact is probably very hard
> > Namely, to make your "symbolic reasons" symbolic in any genuine way

> p35 19. ".. hesitate..." But you already presented it (with out caveat) earlier. Thus you present it twice.

> p35 l18. Not clear why /1000 guarantees <1.
> > Since you can't control the number of reasons (true?) then you can't control the upper bound of //...//

> p36 Ratings for A, F, C not discussed, though this was the natural place

> p36 l6. Formula does not incorporate constraint #3, as far as I can see

> p46 l4. Is there a rule missing (quickee statement) analogously to p42?

> p48 l13. Here's that odd bit about user-PR again

> p51 fn35. The issue of rippling was mentioned at beginning. However, if it is worth discussing, then this feature certainly is integral to its virt)ues and should be disussed in text

> p54-56. I don't see that the first 3 reasons have anything to do with the fixity of facets
> > They would work ok with a growing set of facets
> > The "small reasons" on p56 are the real ones

> p101 l6. It is rather odd to wish to pattern these concepts after those of children when the mode of reasoning is so totally different
> > Ie, AM engages in fantastic feats of disciplined symbolic activity, whereas the 4 year old doesn't even come close

> p102 l-6 What "material"? Referent unclear

> p109 l-3. "best run"? It said it was a "better-than-average" run (p102)

> p111 l-5. Say a word about the distribution of values
> > Or saying they were all set to 200 is sort of meaningless
> > Suppose they all ranged from 199-201 initially

> p112 l-4. "... delay factor still decreased ..." Unclear. Almost like sense reversed. Increased?

> p115 l7. This experiment test the wrong idea
> > Tests: The nbr of reasons is a good score
> > > Compared to the variable scoring
> > Thus it runs afoul of the bad features of "nbr of reasons"

and does not reveal whether full scoring is needed
> How about a lexico graphic ordering on reasons, or dividing reasons into good reasons and bad ones
> 1 good reason wins

> p116 l2. I like "directionality" better than "sophistication"

> p116 l6. I hate to see sections started with pronouns.
> I find it difficulty to find antecedents in section titles

> p117 l-14. You never mentioned the Y-axis before
> Which is very odd in itself
> Is it not true that the concept holds for all pairs of triangles which have two sides co-linear
> Providing that I select the Y-axis right
> Which was probably happening inadvertantly be means of how examples were being constructed

> p118. Am I right -- you advertized 5 experiments and delivered on four?

> p119 l-2. I feel that a comparison with SAM and Bledsoe's program would have been quite useful
> They are extremely close to AM in spirit
> As you see, I think the fact that they are "theorem provers" and AM is "not" is slightly irrelevant

> p120 fn. Earlier you implied the //...// implied normalization

> p122. Here I recalled that you had promised to show that AM ran out of gas -- did that demonstration really happen?

> p122 l-21. I find myself disagreeing with these paragraphs
> The grain of comparison with a mathematician is all important
> You have made no assessment (even conjecture) how many silly things pass through the mathematicians mind when he stares at his work sheet and chew the end of his pencil for a half a minute
> You do have a partial handle on the problem because of the amount of effort spent on foolish concepts
> But here, as you observe, the record isn't all that bad
> AM spent little time on foolish concepts
> Why don't you compute the amount of time spent on good vs bad concepts as well as the number of good vs bad ones
> Given in part on p124, but a time measure rather that a task count might have been better

> p127 7.1.7. I am slightly suspicious of protestions about how easy experiments are when you haven't done and report any great number
> Say 30 or so, since they take only a few minutes, etc.

> p128. <Still to do... Not worth it

> p129 3. I am made uncomfortable with this whole section on education
> It is as if you couldn't wait for anything solid to show up on education, but had to extrapolate like hell
> There is indeed much that is novel in AM, which simply means

that it is highly unstable conceptually

> p131 l7. Your data base looks pretty big to me, but it hardly looks massive
> You live in the same old core space that everyone is filling up
> Many TPer have filled up as much space with clauses as you have with concepts
> Why does that make yours massive and theirs not?

> p133 l-13. Data on this would be nice to have and probably easy to get
> Ie, on the number of slots filled it
> There was one number earlier on original vs all the others, I think, but might be able to do better
> I would also like data on the uses of the different heuristics
> Especially as classified in some way, eg, as mentioned here

> p243. Very odd that you won't even give a proper reference to Ramanujan's paper
> Ditto (but more minor) on Hardy's obit notice

> Bibliography
> p309 Seems to be a flavor on reporting on your development rather than on the papers that you needed to reference
> Ie, in general I would say leave out anything that you didn't need to reference in text
> And if that hurts, then ask why it was so unimportant that there was not need to reference it!
> Don't separate books and articles
> I can't think of a more irrelevant criterion
> Having them all in one list by author makes it easy to find
> By the way, I noticed a number of places where you felt you could reference things in text by name or system names and give not references
> Better to be more compulsive on that
> One example I recall was the discussion of Hearsay
> Isn't Klerner = Klehrer?

Cordell, here is a note I received more recently from Newell:

∂23-JUL-76 1551 FTP:ALLEN NEWELL(A310AN02) at CMUA note
From: ALLEN NEWELL(A310AN02) at CMUA
Date: 23 Jul 1976 1852 EDT
Subject: note
To: DBL at SU-AI
- - - -
Doug: SU-AI has been down for some time to the Net. Finally, realized it and sent you the thesis comments at SUMMEX-AIM.

Didn't mention it, but it occurred to me that Cordell should have a copy, seeing as he's the advisor and I am doing this partially as an official function.

It also occurred to me afterward that you might be a little non-plussed by the critical attitude I took on the thesis (though I did insert a warning at the front). We can talk about all the good things in it when you get here. I am indeed enthusiastic about the effort and think it represents real progress. You will also find, however, that I believe in being highly critical of what my friends do, since I feel that is the only way to make progress. (Pardon -- that is a necessary way to make progress. It surely isn't sufficient.)

a.n.
